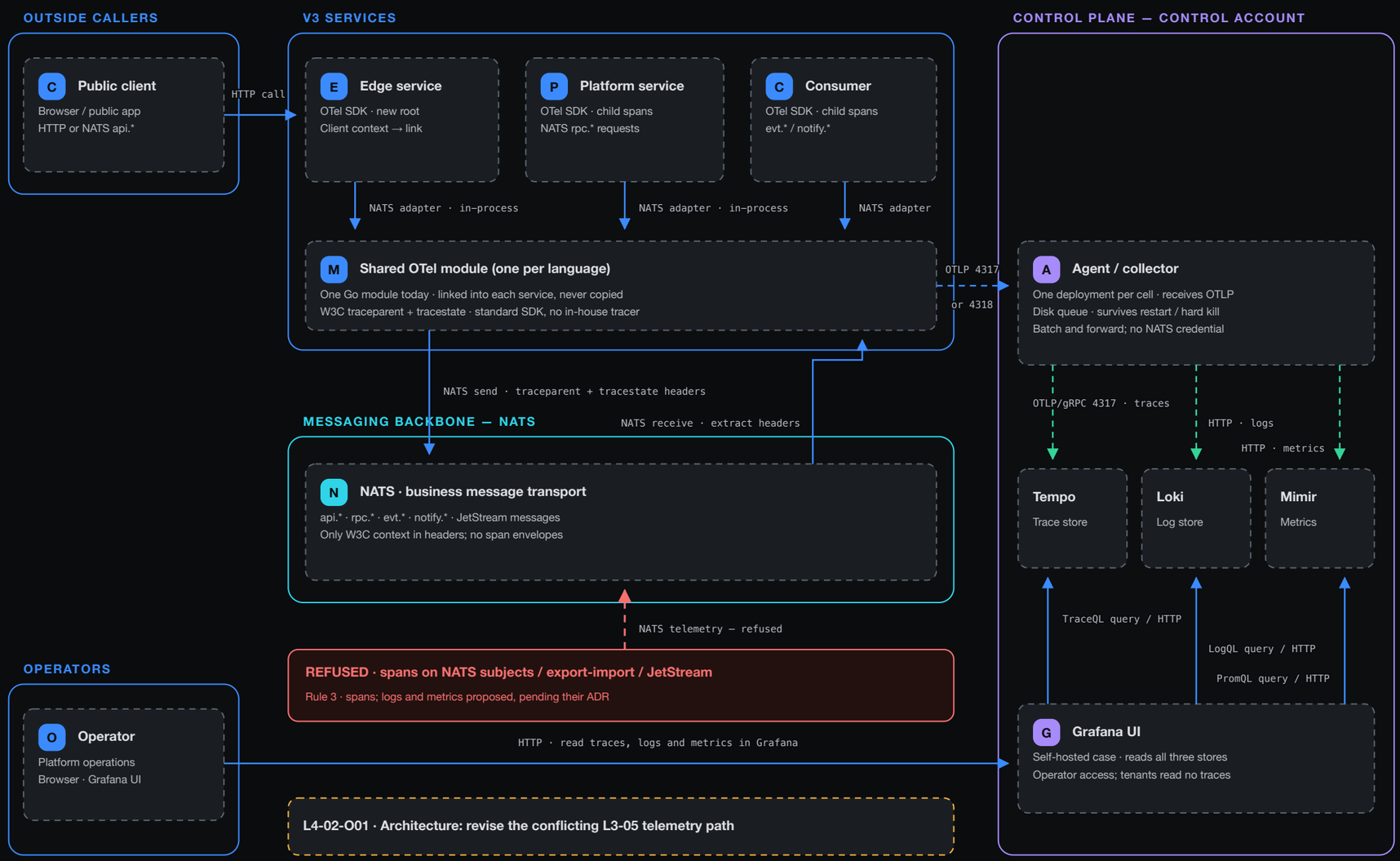
Proposed Solution

Which parts make up the tracing pipeline, and which path do spans take? Defers: logs and metrics rules; retention and sizing (LB-V3-L3-12); wider observability (LB-V3-L3-10). **This is a directional document:** nothing here is built; decision of record ADR-058.



Separate business and telemetry paths. The CONTROL account states ownership, not transport.

Telemetry has its own path. This sheet shows the self-hosted Grafana case; hosting remains open on sheet 5. Account names denote roles, not account identifier values.

SCOPE	**One outside call, one trace.** A span is one timed step. Every V3 service uses its language's standard OTel SDK and W3C Trace Context. `L4-02-001` `L4-02-037`
MODULE	**One module per language.** Today one Go module supplies the platform glue. Services call it; they do not copy it. `L4-02-002`
TRANSPORT	**Service → own-cell agent.** OTLP/gRPC 4317 is the default; OTLP/HTTP 4318 is allowed. The network path is independent of NATS. `L4-02-018` `L4-02-019`
DURABILITY	**Disk buffering is required.** Size for a stated store outage; prove restart and hard-kill survival, full-queue loss counting and complete drain into the store. `L4-02-021`
STORES	**One operator UI.** Tempo keeps traces, Loki logs, Mimir or Prometheus metrics. Tenants read no traces by default. Signal rules beyond tracing are deferred. `L4-02-022` `L4-02-025`
OPEN	**Architecture · L4-02-O01.** Revise LB-V3-L3-05 to remove its telemetry-over-NATS path. ADR-058 rule 3 governs this drawing. `L4-02-019` `L4-02-001`

LEGEND · NOTATION AND STATUS

- **Proposed** — dashed grey tile; nothing built
- **Purple** — control-plane ownership
- **Red** — refused telemetry path
- **Open** — dashed amber; owner + item ID
- **Solid arrow** — synchronous request or query
- **Blue** — services, callers and synchronous calls
- **Dashed blue arrow** — asynchronous service export
- **Cyan** — cell, account and messaging boundaries
- **Dashed green arrow** — store ingest

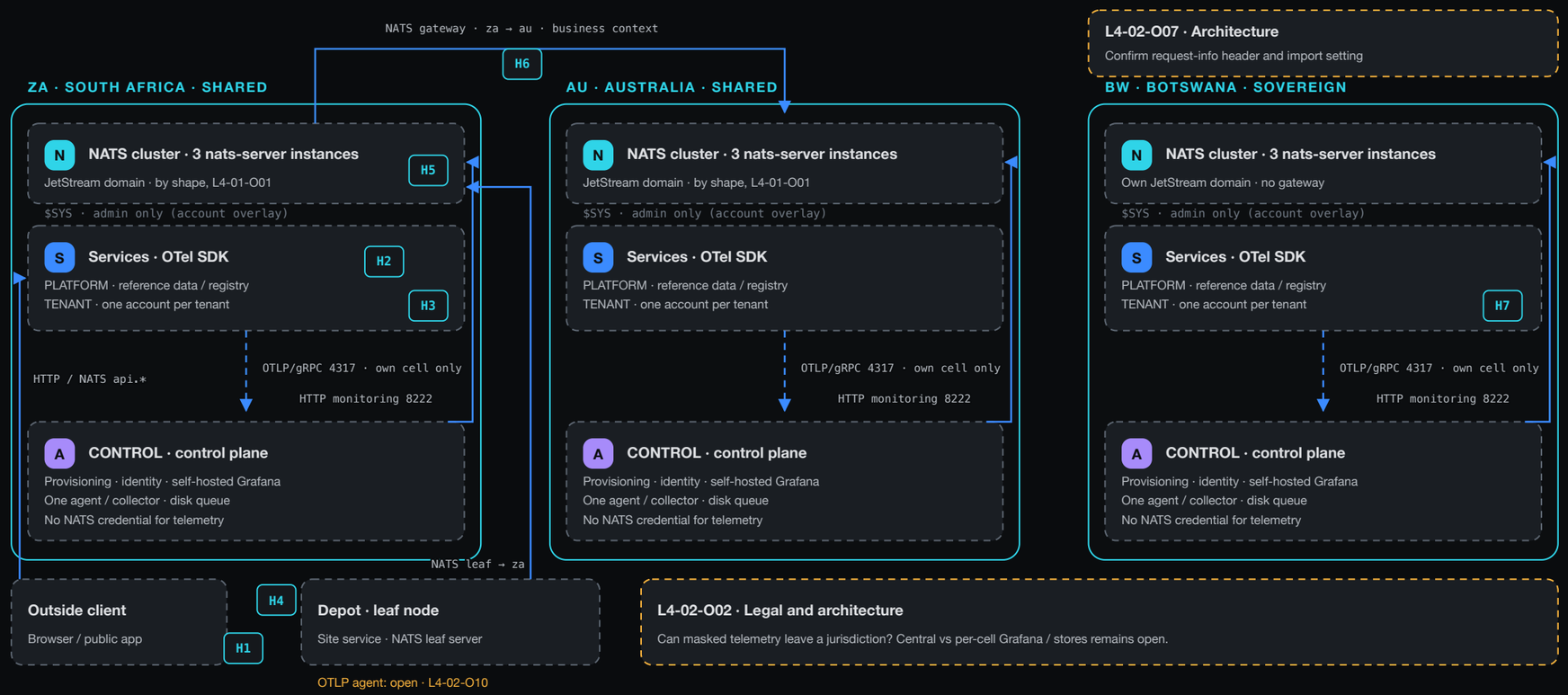
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W3C World Wide Web Consortium
UI User Interface
TraceQL Tempo trace query language
LogQL Loki log query language
PromQL Prometheus query language
ID identifier
api / rpc / evt application interface / remote procedure call / event subject prefixes

Traceability: Parent LB-V3-L3-10 Observability · Decision ADR-058 · Register Proposed-Linebooker-V3-L4-02-Requirements.md

Tracing in the NATS Topology

Where does tracing sit among cells, clusters and accounts, and what does each hop record? Defers: cluster sizing; sovereign connectivity policy; central versus per-cell stores (L4-02-O02). **This is a directional document:** nothing here is built; decision of record ADR-058.



Hop	Crosses	Context carried how	What the span records (attributes that change)
H1	Outside → edge	HTTP / NATS api.* · link only	New platform-sampled root; client cannot join a trace or turn sampling off.
H2	TENANT → PLATFORM	NATS service import · W3C headers	Receiving account from connection; tenant origin from server-written request-info.
H3	One cluster / account	NATS rpc.* / notify.* · W3C headers	Caller and callee may differ in nats.server.name; normal parent / child.
H4	Depot leaf → cell	NATS leaf link · W3C headers	Site nats.server.name = leaf; nats.jetstream.domain = addressed call domain.
H5	Publish → later consume	NATS JetStream · stored headers	Consumer extracts stored context; child span remains in the same trace.
H6	za → au gateway	NATS gateway · W3C headers	cloud.region and nats.cluster.name differ; each side exports to its own agent.
H7	Within sovereign bw	NATS local only · W3C headers	No NATS gateway path out; spans go to the bw agent. Egress: L4-02-O02.

H1–H7 are scenario markers, not a packet itinerary. H2 is the account crossing, H3 an intra-account service call, and H5 a stream within the marked cluster. Gateway direction shown is the H6 scenario.

PLACEMENT — **Account overlays are not regions.** Each cell has $SYS, CONTROL, PLATFORM and tenant accounts. Agent and self-hosted Grafana belong to CONTROL; telemetry needs no NATS credential. `L4-02-020` `L4-02-023` `L4-02-037`

EDGE — **H1 breaks client parentage.** The platform samples a new root; client context is only a link. H2–H6 retain normal parent/child propagation. `L4-02-006` `L4-02-007` `L4-02-026`

TRUST — **H2 is a service import.** The server writes `Nats-Request-Info` on the import; `share` adds requester detail. Tenant comes from the platform's account-to-tenant record. Receiving account and tenant origin stay separate; missing origin is unattributed. `L4-02-013` `L4-02-015` `L4-02-027` **Architecture · L4-02-O07:** confirm header name and import setting against the chosen NATS version.

LOCATION — **Read the live connection at span start.** A reconnect can change nats.cluster.name while cloud.region stays fixed. H3 may change server; H4 records the addressed JetStream domain, omitted if none. `L4-02-010` `L4-02-011` `L4-02-028` `L4-02-029`

HOPS — **H5 retains stored context.** H6 changes region and cluster across the gateway. H7 has no NATS gateway path out of Botswana. `L4-02-008` `L4-02-030` `L4-02-031` `L4-02-032`

RULE — **Monitoring is HTTP, not NATS.** Each agent reads each server's health on 8222. Spans never use subjects, account export/import or JetStream (rule 3). Logs and metrics: proposed the same, pending their ADR. `L4-02-019` `L4-02-024` Jurisdiction remains open: `L4-02-002`

LEGEND · NOTATION AND STATUS

- **Proposed** — dashed grey tile; nothing built
- **Purple** — control-plane agent
- **Cyan badges** — hop scenario locations
- **Open** — dashed amber; owner + item ID
- **Solid blue arrow** — business call / health read
- **Blue** — services, callers and synchronous calls
- **Dashed blue arrow** — asynchronous OTLP export
- **Cyan** — cell, account and messaging boundaries
- **Grey rules** — table structure

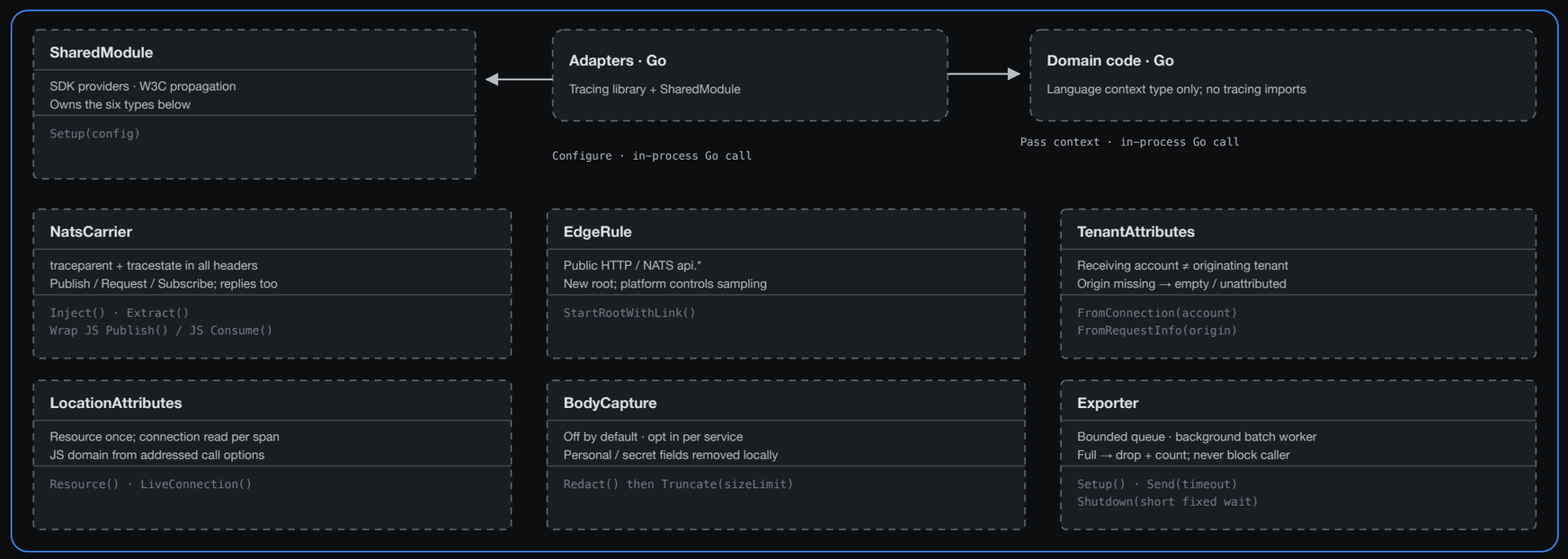
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$SYS NATS system account
H1–H7 numbered hop scenarios
ID identifier
api / rpc / evt application interface / remote procedure call / event subject prefixes

Shared Module and Span Contract

What does the shared module own, and which attributes does every span carry? Defers: package source selection (L4-02-O09), final tenant attribute names (L4-02-O08), and logs / metrics contracts. **This is a directional document:** nothing here is built; decision of record ADR-058.

SHARED MODULE · PROPOSED TYPES · ONE GO MODULE TODAY



Attribute	Set on	Source	Trusted?
`service.name · service.version`	Resource / every span	Deployment configuration; once per process	Yes · deployment
`service.instance.id`	Resource / every span	Process instance metadata; once per process	Yes · deployment
`deployment.environment.name` `cloud.region`	Resource / every span	Deployment configuration; reconnect leaves region fixed	Yes · deployment
`messaging.system` `messaging.destination.name`	Every NATS span	nats; actual message subject	Transport fact; not identity
`messaging.operation.type`	Every NATS span	send / receive / process; kind producer / consumer / client / server	Adapter operation
`nats.server.name · nats.cluster.name` `server.address`	Every NATS span	Live connection at span start, never cached at startup	Yes · live connection
`nats.jetstream.domain`	JetStream spans only	Addressed domain in call options; omit when none named	Yes · service call options
`linebooker.account.receiving`	Receiving service span	Account of the connection on which message arrived	Yes · not tenant identity
`linebooker.tenant.origin`	Receiving service span	Server-written routing / request-info, else empty	Yes if server-written

L4-02-O08 · Architecture: confirm both linebooker.* names. L4-02-O09 · Architecture: confirm shared Go module source.

Class compartments: name, fields / responsibilities, methods. Containment expresses module ownership; grey arrows are in-process Go dependencies, not network traffic. Attribute keys appear without credential, endpoint or account identifier values.

CONTRACT **Standard SDK and W3C context.** The shared module owns these six types. Reuse one module per language. `L4-02-001` `L4-02-002` `L4-02-003` `L4-02-037`

CARRIER **Every inbound and outbound message.** Inject and extract on publish, request, reply, subscribe, JetStream publish and consume. Only adapters and the module import tracing libraries. `L4-02-004` `L4-02-005`

LOCATION **Resource versus live connection.** Resource attributes are present on every span. Re-read server, cluster and address at span start. JetStream domain means the call target, not the connected server's domain. `L4-02-009` `L4-02-010` `L4-02-011`

MESSAGING **Record the actual operation.** messaging.system is nats; destination is the subject; operation is send, receive or process. Select the corresponding producer, consumer, client or server span kind. `L4-02-012`

TRUST **Metadata, never a security boundary.** Receiving account and tenant origin never come from a client header, body or requester field. A PLATFORM or CONTROL connection alone cannot name the tenant. `L4-02-013` `L4-02-014` `L4-02-015` `L4-02-008`

PRIVACY **Body capture is off.** If enabled for a service, redact personal and secret fields, then truncate before export. `L4-02-016`

EXPORT **Bounded and asynchronous.** Queue full means drop and count, never wait. Batch sends have time limits; shutdown waits a short fixed time. `L4-02-017` Module origin remains `L4-02-009`.

LEGEND · NOTATION AND STATUS

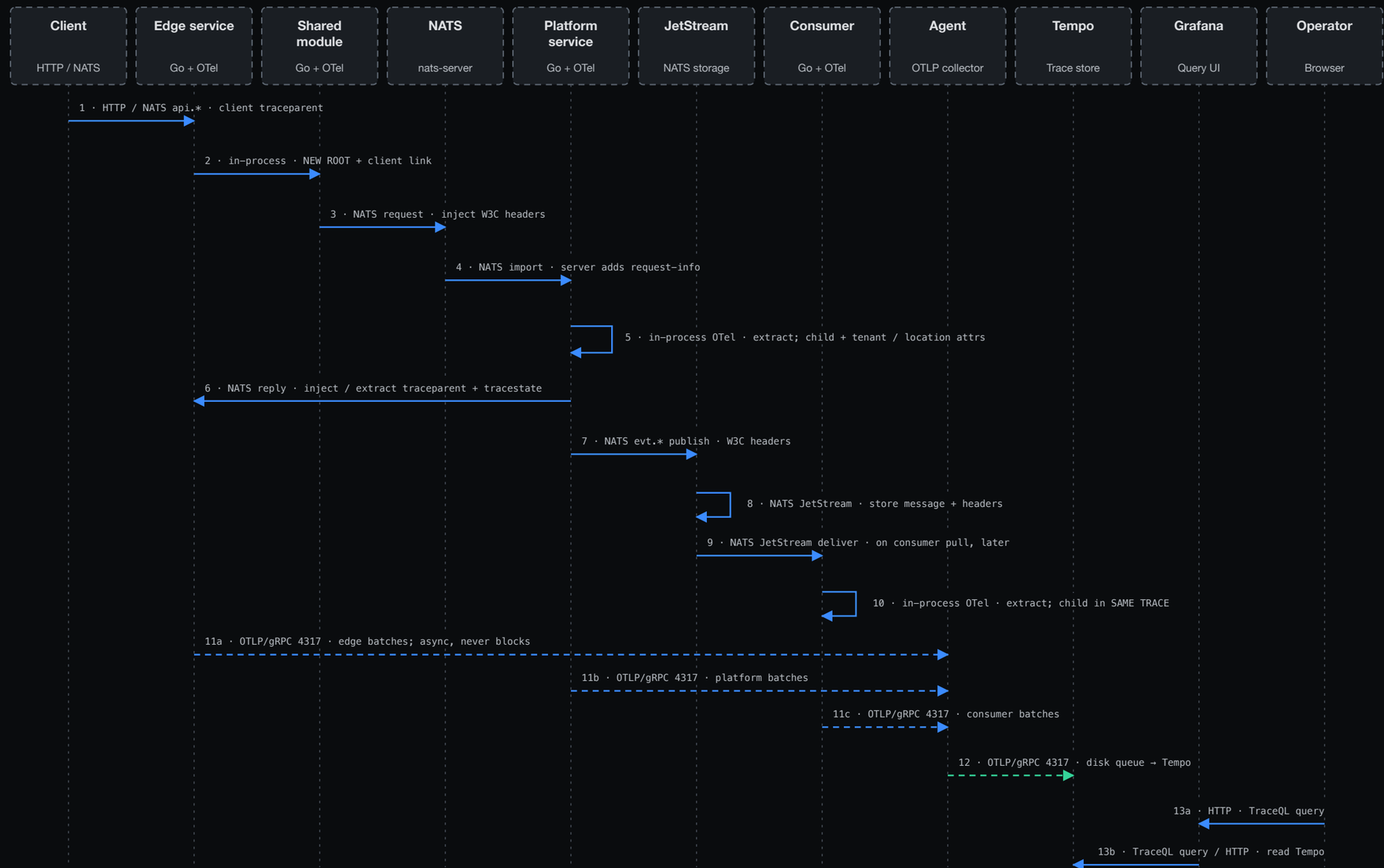
- **Proposed** — dashed grey tile; nothing built
- **Blue boundary** — shared module ownership
- **Open** — dashed amber; owner + item ID
- **Grey arrow** — in-process dependency
- **Blue** — services, callers and synchronous calls
- **Grey rules** — class compartments and table rows
- **Cyan** — cell, account and messaging boundaries
- **Amber text** — proposed names awaiting confirmation

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JS JetStream
UML Unified Modeling Language
ID identifier
api / rpc / evt application interface / remote procedure call / event subject prefixes

One Call, End to End

In what order do the client, services, NATS, agent and Grafana act for one outside call? Defers: business payloads, retry policy and exact timing; sampling rate (L4-02-O05). **This is a directional document:** nothing here is built; decision of record ADR-058.



Dashed arrows are asynchronous. Step 11 is shown below the business interactions for readability; each service exports independently as spans finish. Vertical dotted lines indicate participant lifetime, not a transport.

ORDER — **One call, including later work.** The numbered sequence shows the eleven participants and the required 13 interactions; 11 and 13 expand into distinct one-way arrows. `L4-02-033` `L4-02-037`

EDGE — **1–2: do not trust client parentage.** The shared module starts a new platform-sampled root. The client traceparent becomes a span link only; it cannot disable sampling. `L4-02-006`

HEADERS — **3–6: carry context, not spans.** Inject traceparent and tracestate on outbound request and reply; extract inbound context. The server supplies request-info at the account import. Parent / child applies internally. `L4-02-004` `L4-02-007` `L4-02-015`

EVENT — **7–10: time does not break the trace.** JetStream stores the headers with the message. A later pull consumer extracts those headers and starts a child in the same trace. `L4-02-004` `L4-02-008`

ASYNC — **11: exports are independent of this timeline.** Each finished span enters its service's bounded queue. A background worker batches; full queues drop and count. Send timeout and short shutdown wait bound the work. `L4-02-017`

READ — **12–13: buffer, store, then inspect.** The agent's durable disk queue forwards traces to Tempo; the operator enters a TraceQL query in Grafana, which reads Tempo. No telemetry uses NATS. `L4-02-021` `L4-02-022` `L4-02-025`

LEGEND · NOTATION AND STATUS

- **Proposed** — dashed grey tile; nothing built
- **Solid blue arrow** — synchronous message / local action / query
- **Self arrow** — in-process action; protocol stated
- **Open** — dashed amber; owner + item ID
- **Dashed blue arrow** — nonblocking batch export
- **Blue** — services, callers and synchronous calls
- **Dashed green arrow** — asynchronous store ingest
- **Cyan** — cell, account and messaging boundaries
- **Dotted grey line** — participant lifeline

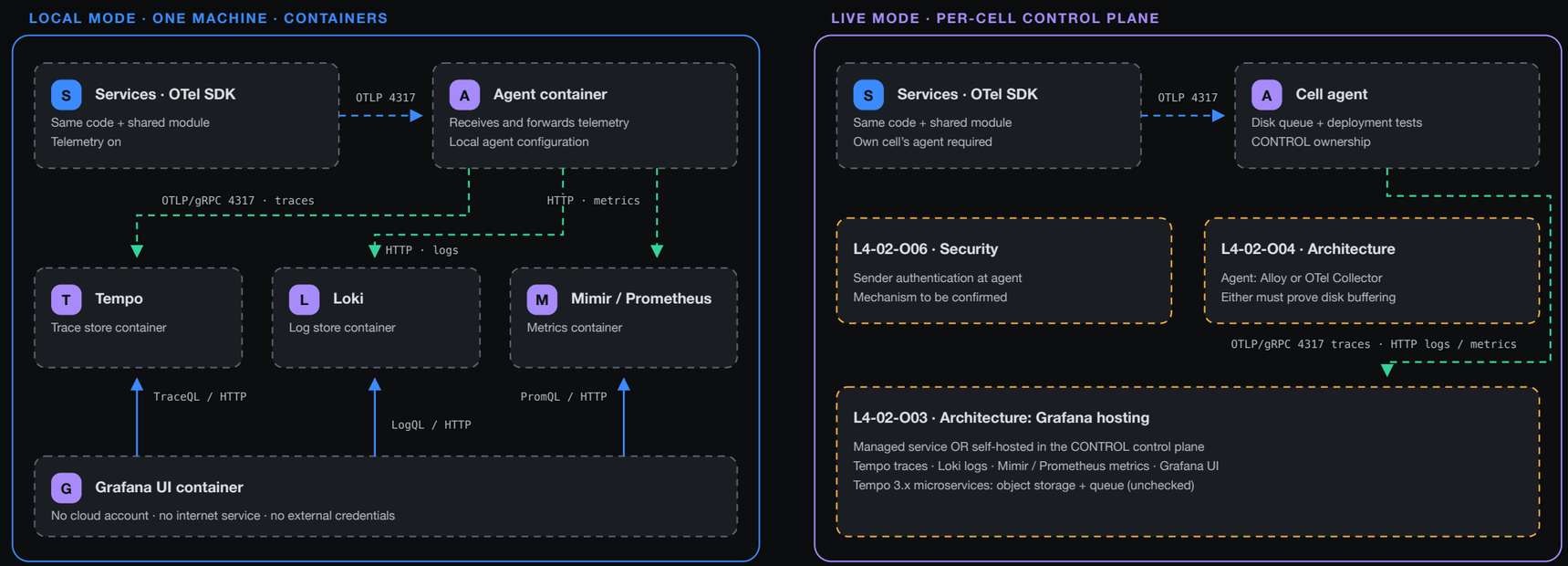
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Traceability: Parent LB-V3-L3-10 Observability · Decision ADR-058 · Register Proposed-Linebooker-V3-L4-02-Requirements.md

Deployment Modes

How does the same code run locally and live, and what changes between them? Defers: agent product, hosting, sender authentication and live sampling; capacity and retention (LB-V3-L3-12). **This is a directional document:** nothing here is built; decision of record ADR-058.



Setting	Local	Live
OTEL_EXPORTER_OTLP_ENDPOINT	Local agent container	Cell agent; missing = startup error; no local fallback
OTEL_EXPORTER_OTLP_PROTOCOL	grpc	grpc (OTLP/HTTP 4318 also allowed)
OTEL_SERVICE_NAME	Service identity from configuration	Same service identity; deployment configured
OTEL_RESOURCE_ATTRIBUTES	deployment.environment.name=local cloud.region from local configuration	deployment.environment.name=live cloud.region from deployment
Platform mode setting	local	live; exact setting name to be confirmed
Body capture	off by default	off by default; per-service opt in
Sampling	Always on	Open L4-02-O05 · Architecture and operations

L4-02-O05 · Architecture and operations: confirm production sampling rate and where it is applied

Configuration is the switch. Values shown are symbolic roles and mode names, never endpoint addresses or credentials. Self-hosted production store requirements must be confirmed against the selected version and mode.

MODES **Two modes, one code path.** Local mode is fully contained on one machine; tracing is on. Live mode changes configuration, never instrumentation code. L4-02-034 L4-02-035 L4-02-037

TRANSPORT **Own-cell OTLP only.** Default gRPC 4317; HTTP 4318 is allowed. No NATS credential carries telemetry. Agent and self-hosted Grafana are control-plane services. L4-02-018 L4-02-023

STARTUP **Fail clearly in live mode.** The module reads the standard OTEL variables plus platform mode. Missing live agent endpoint is a startup error, never a local fallback. L4-02-035 L4-02-036

PROVE **Chosen deployment must pass disk-queue tests.** Restart and hard kill preserve the queue; it holds the stated store outage volume; full queue loss is counted; the complete queue drains with every span stored. L4-02-021

OPEN **Independent choices.** Architecture chooses managed versus self-hosted Grafana and Alloy versus Collector. Security owns sender authentication. Architecture and operations own live sampling. L4-02-003 L4-02-004 L4-02-005 L4-02-006

CAPACITY **Size queue and store ingest together.** Queue capacity, outage duration and retention belong to LB-V3-L3-12 and a later L4. Residency may require per-cell stores. L4-02-021 L4-02-002

LEGEND · NOTATION AND STATUS

- **Proposed** — dashed grey tile; nothing built
- **Purple** — control-plane telemetry services
- **Grey rules** — configuration table rows
- **Open** — dashed amber; owner + item ID
- **Solid blue arrow** — store query
- **Blue** — services, callers and synchronous calls
- **Dashed blue arrow** — service export
- **Cyan** — cell, account and messaging boundaries
- **Dashed green arrow** — store ingest

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